

Displacement Velocity, Acceleration Diagrams

Displacement, Velocity, Acceleration Diagrams when the follower moves with uniform velocity

- Since the follower moves with uniform velocity during its rise and return stroke, the slope of the displacement curve must be constant
- The acceleration or retardation of the follower at the beginning and at the end of each stroke is infinite. This is due to the fact that the is required to start from rest and has to gain a velocity within no time.

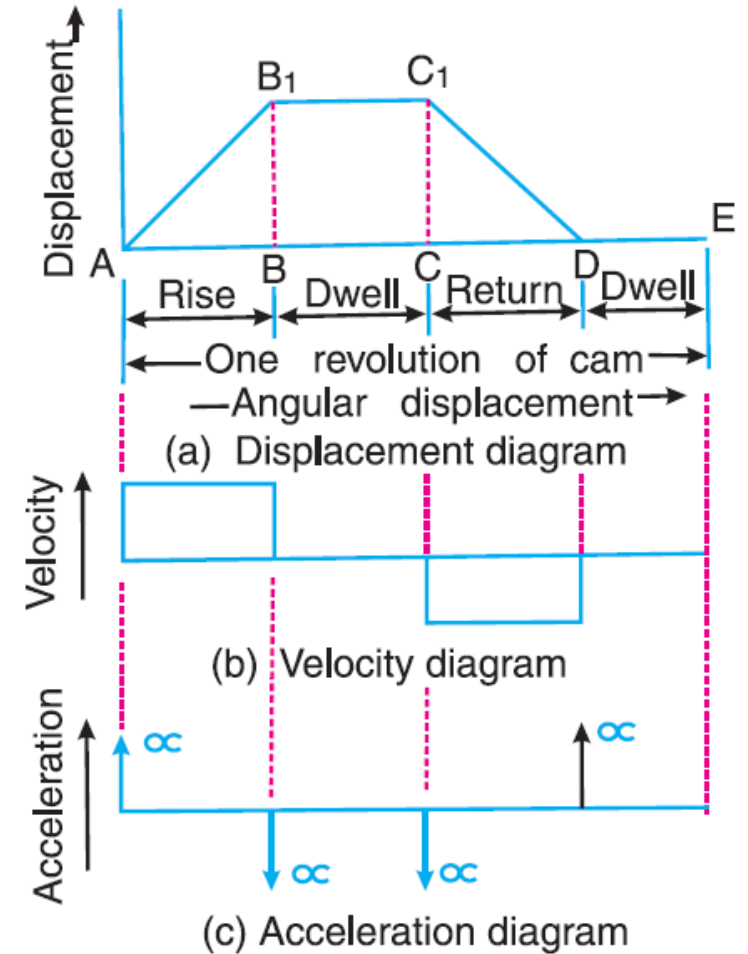


Fig. 20.4. Displacement, velocity and acceleration diagrams when the follower moves with uniform velocity.

Displacement, Velocity, Acceleration Diagrams when the follower moves with SHM

- Follower moves with SHM
- Since the follower moves with a simple harmonic motion, therefore velocity diagram consists of a sine curve and the acceleration diagram is a cosine curve.
- the velocity of the follower is zero at the beginning and at the end of its stroke and increases gradually to a maximum at mid-stroke.
- On the other hand, the acceleration of the follower is maximum at the beginning and at the ends of the stroke and diminishes to zero at mid-stroke.

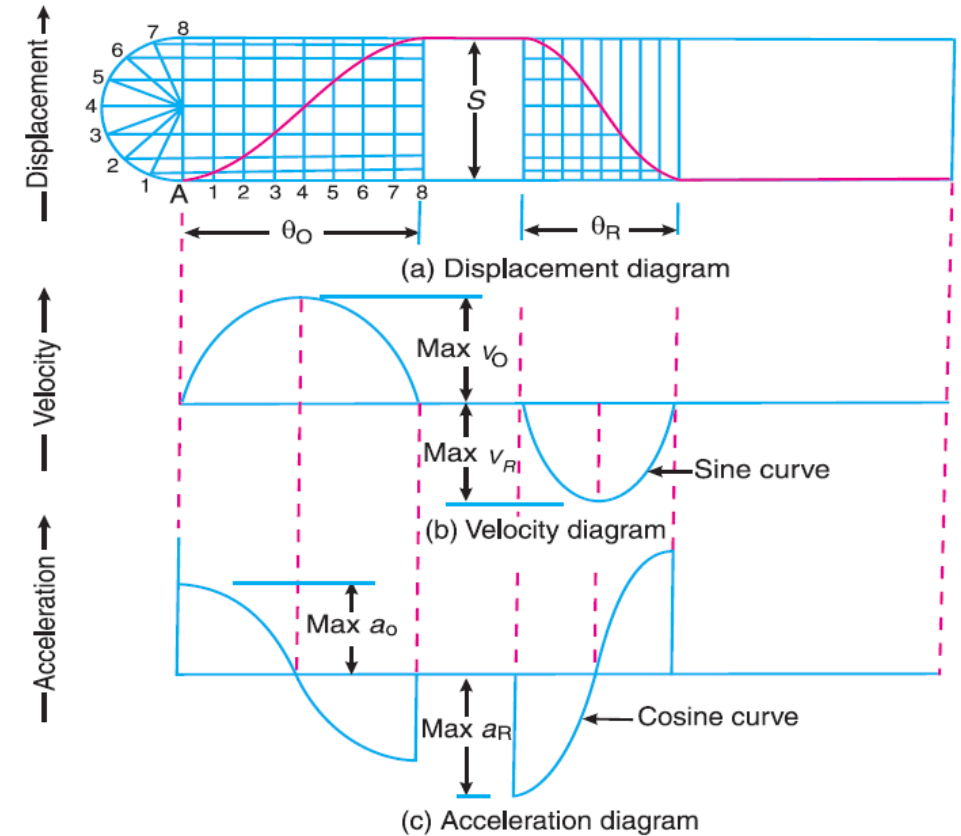


Fig. 20.6. Displacement, velocity and acceleration diagrams when the follower moves with simple harmonic motion.

Let S = Stroke of the follower,

θ_O and θ_R = Angular displacement of the cam during out stroke and return stroke of the follower respectively, in radians, and

ω = Angular velocity of the cam in rad/s.

Time required for the out stroke of the follower in seconds,

$$t_O = \theta_O / \omega$$

maximum velocity of the follower on the outstroke,

$$v_O = \frac{\pi \omega S}{2 \theta_O}$$

Maximum acceleration of the follower on the outstroke,

$$a_O = \frac{\pi^2 \omega^2 . S}{2 (\theta_O)^2}$$

maximum velocity of the follower on the return stroke,

$$v_R = \frac{\pi \omega . S}{2 \theta_R}$$

maximum acceleration of the follower on the return stroke,

$$a_R = \frac{\pi^2 \omega^2 . S}{2 (\theta_R)^2}$$

Displacement, Velocity, Acceleration Diagrams when the follower moves with uniform acceleration and retardation

- Maximum Velocity of the follower during Outstroke

$$v_O = \frac{2S}{t_O} = \frac{2\omega S}{\theta_O}$$

- Maximum Velocity of the follower during Return stroke

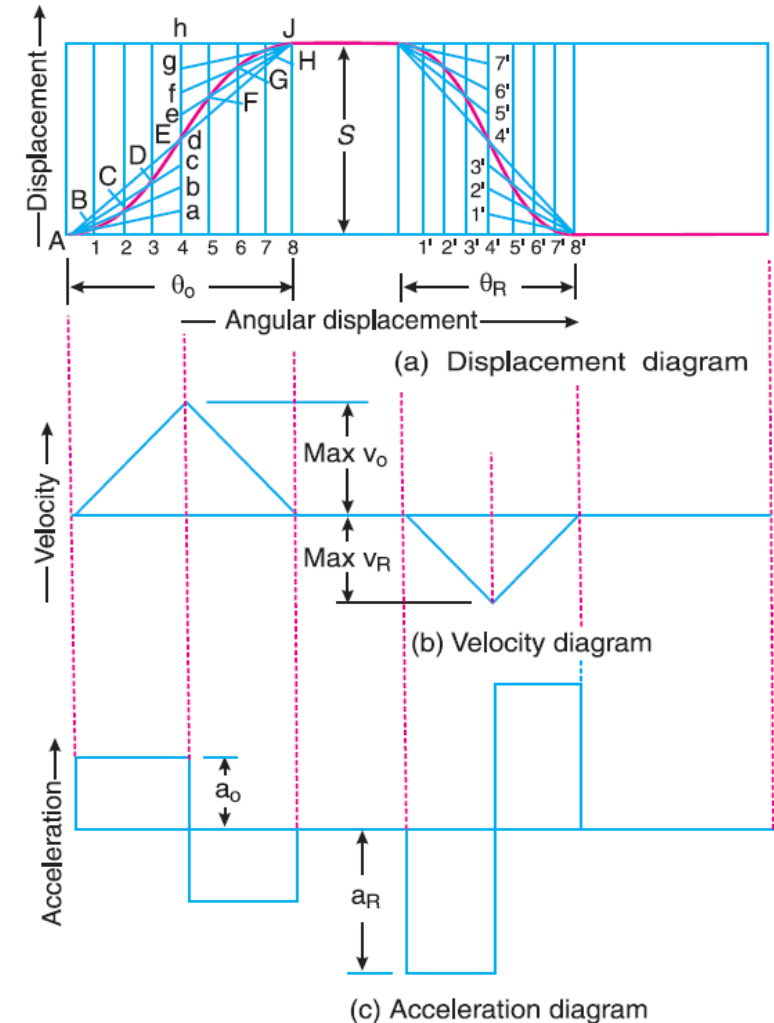
$$v_R = \frac{2\omega S}{\theta_R}$$

- Mean Velocity is the half of maximum velocity
- Maximum acceleration of the follower during outstroke

$$a_O = \frac{4\omega^2 S}{(\theta_O)^2}$$

- Maximum acceleration of the follower during Return stroke

$$a_R = \frac{4\omega^2 S}{(\theta_R)^2}$$



Displacement, Velocity, Acceleration Diagrams when the follower moves with Cycloidal motion

Maximum velocity of the follower during outstroke

$$v_O = \frac{2\omega.S}{\theta_O}$$

Maximum velocity of the follower during Return stroke

$$v_R = \frac{2\omega.S}{\theta_R}$$

Maximum acceleration of the follower during outstroke

$$a_O = \frac{2\pi\omega^2.S}{(\theta_O)^2}$$

Maximum acceleration of the follower during Return stroke

$$a_R = \frac{2\pi\omega^2.S}{(\theta_R)^2}$$

